

Layered Rewrite Draft

Information Processing with Double Entry in the Quantum Era

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Abstract

Digital organizations depend on systems that record, transform, and communicate information. Double-entry accounting is one of the oldest and most durable of those systems. This chapter argues that its durability is not accidental: double entry is a relational information architecture. Its value lies not only in recording balances, but in preserving structured relationships among accounts, statements, and economic events. Quantum information theory provides a useful language for seeing why such relationships matter. In a double-entry representation, account types can play the role of basis states, journal entries can play the role of transformations, and articulated statements can represent information that is not recoverable from isolated accounts. The chapter reviews three stylized settings. First, double-entry structures can represent basic quantum-information operations, including a universal gate set, when amplitude information is tracked separately. Second, articulated account structures can increase channel capacity by allowing more messages to be distinguished than a single isolated account can carry. Third, entanglement-like relational structures improve coordination in the odd-cycle game and suggest a way to think about joint production and performance measurement. The central conclusion is cautious but substantial: double entry does not physically perform quantum computation, but it provides a powerful classical language for representing relational information.

The chapter is organized in layers: the main text gives the conceptual argument, while formal notes and the appendix state assumptions, proposition-level claims, and proof sketches for readers who want to check the logic more closely.

1. Why Double Entry Is An Information Architecture

Organizations have been reshaped by digital transformation. Accounting systems, supply chains, contracting systems, performance dashboards, and customer-service platforms now operate through software, databases, and automated decision rules. As the computational environment changes, it is natural to ask whether the information architecture of organizations must also change.

Double-entry accounting has survived repeated technological revolutions. It survived paper ledgers, mechanical tabulators, mainframes, relational databases, enterprise resource planning systems, and cloud platforms. Its persistence suggests that double entry is more than a convenient recording convention. It is a structure for organizing information.

The question for this chapter is whether that structure remains useful as quantum information concepts become more visible in computing and organizational design. Our answer is yes, but the claim must be stated carefully. We do not claim that financial statements are quantum objects. We do not claim that accounting systems physically perform quantum computation. We claim that double entry has a relational structure that can represent important features of information processing, including features that are central in quantum information theory.

The intuition is simple. A single account balance is data. A set of articulated accounts is code. The information in double entry often resides not in one number by itself, but in the relationship among numbers, accounts, and statements.

Consider accounts receivable. Suppose a reader knows beginning accounts receivable, ending accounts receivable, and sales on account. The reader can infer cash collected from customers because the accounts receivable relation must hold:

Beginning accounts receivable + credit sales - cash collections = ending accounts receivable.

Cash collections are not visible from sales alone. They are not visible from the change in receivables alone. They become visible because the reader understands how the two statements and the account relation articulate. The relationship is the code that lets the reader extract information.

Formal layer 1: Articulation as inference code. *Let AR_0 be beginning receivables, AR_1 ending receivables, S credit sales, and C cash collections. The articulation relation $AR_1 = AR_0 + S - C$ implies $C = AR_0 + S - AR_1$. The information recovered by the receiver is not contained in any isolated number; it is recovered from the relationship among the numbers.*

This is the sense in which accounting is an information science. Shannon's theory studies how uncertainty is represented, transmitted, and reduced. Accounting performs a related organizational function. It represents states of the firm, transmits information about events, and permits users to infer what happened from a structured set of signals.

Fellingham, Lin, and Schroeder (2022) develop a broader three-way equivalence among decisions, accounting numbers, and information. In one frame, an economic actor chooses among uncertain outcomes. In a second frame, the same situation is represented through accounting income and book value. In a third frame, it is represented through entropy and information. The details depend on the Kelly decision problem, geometric mean valuation, and assumptions that connect repeated decisions to long-run growth. The point for this chapter is the conceptual one: decision quality, accounting measurement, and information are not separate topics. They are different representations of the same underlying uncertainty.

Formal layer 2: Decision, accounting, and information equivalence. *Under the Kelly long-run growth frame, arbitrage-free pricing, and geometric mean valuation, changes in expected log return can be written as changes in information, commonly measured through entropy. The chapter uses this result as prior literature rather than reproving it; its role here is to justify treating accounting measurement as an information-processing problem.*

This information perspective helps explain why double entry is a promising language for quantum information analogies. Quantum theory is also relational. It represents states, transformations, measurements, and correlations. The most important quantum effects are not always properties of isolated objects; often they arise from relationships among parts of a system. Double-entry accounting, in its own classical domain, is also built around relationships.

The rest of the chapter develops that idea in three stages. Section 2 explains how basic quantum-information objects can be represented with T-accounts and journal entries. Section 3 examines how relational structures analogous to entanglement matter for information transmission, coordination, and performance measurement. Section 4 concludes by returning to the organizational meaning of double entry.

2. Representing Quantum Information With Accounting Objects

Quantum information theory is usually presented with Hilbert spaces, state vectors, unitary operators, tensor products, and measurements. That notation is powerful, but it can be distant from the language of business. Fellingham, Lin, and Schroeder (2025) show that a large part of the basic structure can be represented using familiar accounting objects.

Quantum information concept	Accounting representation	Interpretation
Qubit basis state 0	E account	One basis state of a two-state system
Qubit basis state 1	A account	The other basis state of a two-state system
X gate	Reclassification entry	Switches E and A
Z, S, and T gates	Revaluation entries	Change phase on the A side of the representation
Hadamard gate	Combined reclassification and revaluation, with amplitude tracked separately	Creates a superposition over basis states
Product basis state	Concatenated account string such as EA or AEA	Represents a computational-basis state of a multi-qubit system
Measurement	Outcome probabilities computed from amplitudes	Converts the represented state into an observed basis outcome

The accounting representation is best understood as a notation and a model. It tracks the basis-state structure of a quantum system and the transformations applied to that structure. Complex amplitudes, phases, normalization constants, and measurement probabilities must still be tracked explicitly. This distinction matters. The accounting notation is useful precisely because it gives a concrete way to visualize state labels and transformations, not because it eliminates the mathematics of quantum probability.

***Formal layer 3: Representation convention.** A complete represented state requires two components: account labels for the computational-basis structure and a separately maintained amplitude vector. The account labels identify basis states such as E, A, EA, or AEA. The*

amplitudes determine interference and measurement probabilities. The accounting notation is therefore a structured representation of quantum-information syntax, not a substitute for amplitudes.

2.1 Basis States And Journal Entries

The fundamental unit of quantum information is the qubit. A qubit has two computational basis states, conventionally written as 0 and 1. In the double-entry notation, these are represented by two types of accounts. The account E represents basis state 0. The account A represents basis state 1. Both accounts are treated as unit accounts, so scale is abstracted away.

Quantum gates transform states. The simplest example is the X gate, or bit flip. It changes 0 to 1 and 1 to 0. In accounting notation, this is represented by a reclassification entry that moves the account between E and A.

The Z gate is a phase flip. It leaves the 0 state unchanged and changes the sign of the 1 state. In accounting notation, it is represented by a revaluation entry applied to A. More general phase-shift gates, such as the S gate and T gate, are represented by fractional revaluation entries. These entries trace a structure analogous to movement around the unit circle in the complex plane.

This analogy is useful, but it needs one qualification. In quantum mechanics, an overall global phase is not physically observable. Relative phase becomes meaningful when states are in superposition. The accounting notation tracks phase labels as part of a vector representation; the interpretation of those labels depends on the surrounding state and measurement context.

2.2 Superposition And The Hadamard Gate

A qubit can be in a superposition of its two basis states. The Hadamard gate is the standard operation that creates a balanced superposition. Applied to basis state 0, it produces an equal-amplitude combination of 0 and 1. Applied to basis state 1, it produces an equal-amplitude combination of 0 and negative 1.

In the accounting representation, the Hadamard gate combines the effects of reclassification and revaluation. Acting on a basis account, it produces a stacked representation that records both resulting basis components. The amplitude factor, $1/\sqrt{2}$, is not carried by the T-account label itself; it is recorded separately.

For accounting readers, the important idea is not that financial statements literally enter quantum superposition. The idea is that the notation represents information across more than one linked row or statement. The information-bearing object is no longer one isolated account. It is a structured representation of possible account states together with the amplitudes used to compute measurement probabilities.

2.3 Composite Systems And Entangled States

The treatment of multi-qubit systems requires particular care. In quantum theory, the state space of a composite system is the tensor product of the state spaces of its component systems. Computational-

basis product states can be written as concatenated account strings. For example, EA represents a two-account basis state, and AEA represents a three-account basis state.

But not every state of a composite system is a product state. A general two-qubit state may be a superposition of EE, EA, AE, and AA. Some such states are entangled, meaning they cannot be factored into a state of the first qubit and a state of the second qubit. This correction is essential. If one says that the state of a composite system is simply the tensor product of the states of the parts, one accidentally excludes entanglement, which is exactly the phenomenon of interest.

Conditional operations then become natural. A controlled-NOT gate, or CNOT, flips a target qubit when a control qubit is in state 1. In accounting notation, if A is the control state and E is the target state, a controlled reclassification changes AE into AA. More generally, controlled journal entries let one account transformation depend on the state of another account.

***Formal layer 4: Product and entangled states.** A two-account basis string such as EA is a product basis state. A general two-account state has the form $\alpha_{EE} EE + \alpha_{EA} EA + \alpha_{AE} AE + \alpha_{AA} AA$. It is separable only when the coefficient matrix has rank one. Bell states are not rank-one states, so they cannot be decomposed into independent states of the two accounts.*

2.4 Representational Universality

The Toffoli gate is a three-qubit gate with two controls and one target. It flips the target if and only if both controls are in state 1. It is universal for classical reversible computation. Aharonov (2003), building on Shi (2003), gives a simple proof that Toffoli together with Hadamard forms a universal gate set for quantum computation.

Fellingham, Lin, and Schroeder (2025) provide an accounting construction of the Toffoli gate and show how the Hadamard gate can be represented in the same framework. The safe conclusion is therefore:

The double-entry framework can represent a universal quantum gate set, provided the account notation is paired with explicit amplitude bookkeeping.

This is an important result. It does not mean that ordinary accounting systems are quantum computers. It means that double-entry notation has enough representational structure to describe the primitive operations from which quantum algorithms can be built. That is already a striking observation. A business language built around accounts and journal entries can represent the logic of state transformation, conditional operation, and reversibility that appears in quantum computation.

***Proposition 1: Representational universality.** If the accounting framework can encode the Hadamard gate and the Toffoli gate, and if amplitudes are tracked separately, then the framework can represent a universal gate set for quantum computation. The proposition is representational: it concerns the expressive vocabulary of the accounting model, not physical implementation.*

3. Entanglement, Articulation, And Organizational Information

The most important lesson from quantum information theory is not merely that states can be represented as vectors. It is that relationships among parts of a system can carry information that is not present in the parts separately. In quantum theory, the strongest form of this phenomenon is entanglement. In accounting, the closest analogue is articulation: the disciplined relationship among accounts and statements that lets a reader infer more than any isolated balance reveals.

This section reviews three settings in which relational structure matters.

3.1 Bell States As A Model Of Nonseparable Relationships

A Bell state is a maximally entangled two-qubit state. It can be generated by starting with a two-qubit basis state, applying a Hadamard gate to the first qubit, and then applying a CNOT gate using the first qubit as the control and the second as the target.

In accounting notation, this construction creates a representation in which the two account columns cannot be described independently. The information is in the joint state. Looking at one account alone does not recover the full information-bearing structure.

This is where the analogy becomes powerful. Many organizational relationships are not well described by isolated components. A division may matter because of how it coordinates with another division. A performance measure may matter because of how it links individual actions to joint output. A financial statement may matter because of how it articulates with another statement. Bell states provide a formal model of nonseparability; articulation provides the accounting mechanism by which nonseparable information becomes visible.

3.2 Channel Capacity: Articulation Lets One Account Carry More

The information-property paper studies double entry as a communication channel. A sender has private information and encodes messages as journal entries. A receiver observes the resulting account structure and tries to infer which message was sent.

The model abstracts away from dollar magnitudes so that it can isolate the information carried by structure. This abstraction is important. In practice, magnitudes matter enormously. But by temporarily removing scale, the model asks a narrower question: how much information is carried by the classificatory and articulating structure of double entry itself?

The basic message set has four possibilities:

- I: no entry.
- X: a reclassification entry.
- Z: a revaluation entry.
- X followed by Z: a reclassification followed by a revaluation.

Four distinguishable messages correspond to two bits of information. A single isolated account cannot distinguish all four messages. It achieves only one-bit capacity because some resulting signals are equivalent. The key result is that two-bit capacity becomes possible when the system has linked statement rows and at least two accounts satisfying the articulation identities.

Proposition 2: Two-bit channel capacity. *In the information-property model, one manipulated account can transmit two bits through the message set $\{I, X, Z, XZ\}$ only when it is embedded in an articulated structure that preserves enough relational information to distinguish all four resulting signals. In the accounting interpretation, this requires linked statement rows and at least two accounts satisfying the articulation identities.*

Put less formally: one account can carry more information when it is embedded in an articulated structure. The second account need not itself be manipulated. Its presence matters because it helps preserve the relational code that lets the receiver distinguish among messages.

This is the accounting heart of the chapter. The power of double entry is not "recording everything twice." It is recording events in a structured system of relationships. Those relationships let users infer distinctions that a single-entry scalar record would collapse.

The n-account extension is subject to the same conditions. Under the conditions of the model, an n-account articulated structure can generate n bits of channel capacity when the relevant accounts satisfy the articulation constraints. The capacity scales because the relational structure preserves distinguishable patterns across accounts and statements.

Proposition 3: Scaling condition. *The n-bit extension is not a free consequence of having n labels. It depends on the same articulation condition account by account. The structure scales when each relevant account contributes an independently distinguishable relational pattern.*

3.3 Coordination: The Odd-Cycle Game

The odd-cycle game provides a vivid example of why relational structure can matter. In the three-seat version, Alice and Bob are assigned seats around a round table with seats labeled 0, 1, and 2. The referee gives them either the same seat or adjacent seats. Each player sees only his or her own seat and must report one of two colors, Red or Blue. They win if they report the same color when assigned the same seat and different colors when assigned adjacent seats.

The restriction is that the players cannot simply report their seat numbers. There are three possible seats but only two possible reports. Before the game begins, they may agree on a strategy. After receiving their seat assignments, they cannot communicate.

Classically, the best strategy wins five of the six possible assignments. The reason is intuitive. Trying to two-color an odd cycle must fail somewhere. Around a triangle, at least one adjacent pair must receive the same color, so at least one adjacent-seat query is lost. Thus the classical ceiling is 5/6.

The quantum version changes the informational relationship between the players. If the two players share an entangled Bell state and then choose measurement angles based on their seat assignments, their maximum winning probability rises to

$$1/2 + \sqrt{3}/4 = \cos^2(\pi/12), \text{ approximately } 0.933.$$

This strictly exceeds the classical ceiling of 5/6, approximately 0.833. The value is also consistent with the general odd-cycle quantum limit $\cos^2[\pi/(4n)]$ for n seats.

Proposition 4: Odd-cycle coordination advantage. *In the three-seat odd-cycle game, the best classical strategy wins with probability $5/6$. With the entangled Bell-state strategy used in the companion paper, the winning probability is $1/2 + \sqrt{3}/4 = \cos^2(\pi/12)$. The advantage belongs to this specified informational relationship and strategy space; it does not imply that any use of superposition improves coordination.*

The companion odd-cycle paper compares three cases:

Case	Initial relationship	Winning probability	Interpretation
Basis product state	No superposition and no entanglement	$5/6$	Reproduces the classical benchmark
Bell state	Entangled relational structure	$1/2 + \sqrt{3}/4$	Strictly exceeds the classical benchmark
Superposition without entanglement	One account in superposition but no relational link	$1/2$	Performs worse in this model

The lesson is narrow but important. In this model, superposition alone is not enough. The advantage comes from the entangled relationship between the two players' systems. Translated into accounting language, the result suggests that the value of double entry lies not in isolated accounts and not even in multiple possible account states, but in structured linkages among accounts.

The analogy to single-entry and double-entry systems is useful but limited. The classical game resembles a scalar or single-entry representation because each player reports from an isolated local input. The quantum version resembles a relational double-entry representation because the players' outcomes are linked by a shared structure established before the reports. This is an analogy, not an identity. But it is a useful analogy because it makes the central accounting idea visible: links can be information-bearing.

3.4 Joint Production And Performance Measurement

Organizations use information not only to communicate, but also to coordinate production and design incentives. When two agents work independently, individual performance measures can be highly useful. They help a principal evaluate effort and assign rewards. When production is joint, however, individual measures can miss the structure that creates value.

Demski, Fellingham, Lin, and Schroeder (2008) study a principal-agent setting in which the value of individual and group measurement depends on the degree of productive interaction. We rely on that result here as an organizational analogue rather than as a quantum claim. The conceptual connection is this: when output depends on interaction, a measurement system that separates individuals may destroy or obscure information about joint production. A group measure may be less informative about each agent separately, but more faithful to the relational structure that produces the output.

Formal layer 5: Measurement level. *The agency implication is not that productive agents are quantum-entangled. The formal point is about the level at which information is preserved. If the payoff-relevant object is joint output, then a measurement basis that separates agents may discard information about the joint object, while a group measure may preserve it.*

The quantum analogy is therefore modest. Synergy is not literally quantum entanglement. But entanglement supplies a useful model of nonseparability. It reminds us that, in some systems, the whole is not merely the sum of independent parts. If the economically relevant object is the relationship between agents, divisions, or accounts, then a measurement system that preserves the relationship may dominate one that breaks it apart.

This lesson is important for performance measurement. More measurement is not always better. When tasks are separable, more detailed individual measurement can improve incentives. When tasks are highly interdependent, individual measurement may encourage agents to optimize separate signals rather than joint performance. The design problem is not simply to measure more; it is to measure at the right level of relational structure.

4. What The Quantum Analogy Teaches Accounting

The preceding analysis points to a disciplined conclusion. Quantum information theory does not prove that accounting is quantum. It does not replace accounting theory. It gives accounting researchers a new vocabulary for something accounting has long known: relationships matter.

Double entry is durable because it is relational. It connects resources and claims. It connects income and changes in equity. It connects accrual measures and cash-flow consequences. It connects transactions to statements and statements to one another. A single-entry record can say that cash changed. A double-entry system can say why cash changed, what else changed, and how the event fits within the organization's structure.

The formal models reviewed in this chapter support three conclusions.

First, double-entry notation can represent state transformations with surprising richness. T-accounts and journal entries can encode basis states, reversible transformations, conditional operations, and a universal quantum gate set when amplitudes are tracked separately. This makes the accounting representation pedagogically useful for introducing quantum information concepts to business readers.

Second, articulated accounting structures can increase information capacity. The information-property model shows that linked statements and accounts can distinguish messages that isolated accounts cannot. The result is best understood as a structural result: articulation itself carries information.

Third, relational structure can improve coordination. The odd-cycle game shows, in a simple and memorable setting, how an entangled relationship can outperform any classical isolated reporting strategy. The agency setting suggests a parallel organizational lesson: when production is joint, measurement should preserve the relationship that creates value.

These conclusions offer one information-theoretic explanation for the endurance of double entry. The system has lasted not only because it detects errors or satisfies convention, but because it is an efficient language for relational information. As organizations become more digital, automated, and eventually quantum-aware, that language remains valuable.

The most interesting implication is not that accounting must become quantum. It is that accounting already contains a sophisticated theory of linked states. Quantum information theory helps us see that old architecture with new eyes.

Technical Appendix: Formal Layer

A.1 State Labels And Amplitude Bookkeeping

The accounting notation used in this chapter separates basis labels from amplitudes. The label E represents computational basis state 0, and A represents computational basis state 1. For a single qubit, a general state is not simply E or A, but $\alpha E + \beta A$, with $|\alpha|^2 + |\beta|^2 = 1$. The account label identifies the basis component; the amplitudes α and β determine interference and measurement probabilities.

For a multi-account system, concatenated labels such as EA or AEA identify computational-basis product states. A general two-account state is:

$$\psi = \alpha_{EE} EE + \alpha_{EA} EA + \alpha_{AE} AE + \alpha_{AA} AA.$$

The accounting notation is useful because journal-entry operations can be described on the basis labels. It is incomplete unless the corresponding amplitudes are carried along.

A.2 Product States, Tensor Products, And Bell States

The state space of a composite quantum system is the tensor product of the component state spaces. This statement concerns spaces, not necessarily individual states. Product basis states factor; general states need not.

For example, the Bell state usually written Φ^+ has the form:

$$\Phi^+ = (EE + AA) / \sqrt{2}.$$

If Φ^+ were separable, it could be written as $(aE + bA)(cE + dA)$. Comparing coefficients would require $ac = 1/\sqrt{2}$, $bd = 1/\sqrt{2}$, $ad = 0$, and $bc = 0$. These conditions cannot all hold. The state is therefore nonseparable. This is the formal reason the chapter treats Bell states as models of relational information that cannot be recovered from either component alone.

A.3 Universality Claim

The chapter uses a narrow universality claim. The Toffoli gate is universal for classical reversible computation. Aharonov (2003) and Shi (2003) establish that Toffoli together with Hadamard gives a universal gate set for quantum computation, in the sense that arbitrary quantum operations can be approximated to arbitrary accuracy by circuits built from that set.

The accounting framework does not physically implement those circuits. The claim is that the framework can represent the relevant primitive gates. With explicit amplitude bookkeeping, this gives the framework the representational vocabulary needed to describe universal quantum computation.

A.4 Channel Capacity Conditions

The channel-capacity model uses four messages: I, X, Z, and XZ. Because there are four messages, the maximum possible information conveyed by the message choice is two bits. A single isolated account does not generally produce four distinguishable signals because some outcomes are scalar multiples or otherwise equivalent under the model's signal criterion.

Two-bit capacity arises when the manipulated account is embedded in an articulated structure that preserves enough relational information to distinguish all four messages. In accounting terms, the important conditions are:

- multiple linked statement rows;
- at least two accounts in the structure;
- articulation identities that constrain how account states can vary across rows.

The n-account extension relies on the same logic. Capacity scales when the relational structure preserves independently distinguishable patterns across accounts.

A.5 Odd-Cycle Game Calculation

In the three-seat odd-cycle game, there are six equally likely seat assignments: three same-seat assignments and three adjacent-seat assignments. A classical two-coloring of a three-cycle must fail on at least one adjacent edge. Therefore the best classical strategy can win at most five of six cases, giving probability 5/6.

The entangled strategy changes the measurement relationship between Alice and Bob. In the three-seat case, the optimal entangled winning probability is:

$$1/2 + \sqrt{3}/4 = \cos^2(\pi/12) = \text{approximately } 0.933.$$

The comparison is informative because it separates three cases: no superposition and no entanglement reproduces the classical benchmark; entanglement improves the benchmark; superposition without entanglement performs worse in the model. The result supports a precise conclusion: in this coordination problem, the useful resource is the nonseparable relationship, not mere multiplicity of possible states.

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